

OBSTETRICS

The impact of body mass index on misoprostol dosing for labor induction: a comparison of vaginal and buccal dosage forms



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BACKGROUND: Despite the increasing prevalence of obesity, strategies to manage induction of labor in obese patients remains underinvestigated.

OBJECTIVE: This study aimed to determine whether the number of doses to achieve active labor differed across body mass index categories for buccal misoprostol vs vaginal misoprostol during labor induction.

STUDY DESIGN: This was a secondary analysis of the Induction with MisoPROstol: Oral mucosa versus Vaginal Epithelium study (identification number: NCT 02408315), a triple-blinded, placebo-controlled randomized controlled trial of vaginal misoprostol vs buccal misoprostol in term patients undergoing labor induction. The primary outcome for this analysis was the number of doses of misoprostol needed for active labor (defined as a cervical dilation of at least 6 cm). The participants were grouped into 3 body mass index categories (nonobese: ≤ 29 kg/m²; obese: 30–40 kg/m²; morbidly obese: ≥ 41 kg/m²). Groups were compared using the Wilcoxon signed-rank test and chi-squared or Fisher exact test. Generalized linear models were used to adjust for ethnicity, race, dosing route, and indication for labor induction. Significance was defined as $P < .05$.

RESULTS: Of 299 participants with complete data, 63 (21%) had a body mass index of ≤ 29 kg/m², 165 (55%) had a body mass index between 30 and 40 kg/m², and 71 (23%) had a body mass index of ≥ 41 kg/m². The groups did not differ in age, race, gestational age, nulliparity, initial Bishop score, or route of dosing. There were significantly more non-Hispanic participants in the obese categories ($P = .02$). A higher proportion of participants with obesity or morbid obesity had diabetes mellitus (6.4% in the nonobese group vs 13.9% in the obese group vs 14.1% in the morbidly

obese group) or a hypertensive disorder (7.9% in the nonobese group vs 24.9% in the obese group vs 26.8% in the morbidly obese group) as the primary indication for labor induction ($P = .03$). In the univariate analysis, compared with the nonobese group, more doses of misoprostol were needed to achieve active labor in the obese and morbidly obese groups when dosed buccally (2.1 vs 2.8 vs 2.9, respectively; $P = .01$) but not when dosed vaginally (1.9 vs 2.1 vs 2.3, respectively; $P = .21$). The rates of vaginal delivery in < 24 hours, cesarean delivery, or cesarean delivery for fetal distress did not significantly differ across body mass index category or dosing route. When adjusting for indication for labor induction and self-reported race/ethnicity, buccal dosing required more doses to achieve active labor, regardless of body mass index (24% more expected doses; $P < .005$). Similarly, when controlling for indication for labor induction and self-reported race/ethnicity, obesity was independently associated with requiring more doses for active labor for both dosage routes (body mass index of 30–40 kg/m²: 24% more expected doses; confidence limits of 1.01–2.68; $P < .047$; body mass index of ≥ 41 kg/m²: 28% more expected doses; confidence limits of 0.99–3.09; $P < .05$).

CONCLUSION: Obesity (body mass index of ≥ 30 kg/m²) was associated with the need for more misoprostol doses to achieve active labor. In addition, buccal dosing was independently associated with a higher number of doses than vaginal dosing, especially in patients with obesity.

Key words: body mass index, buccal misoprostol, cervical ripening, induction, labor, labor induction, misoprostol, obesity, vaginal misoprostol

Introduction

Obesity is the most common medical comorbidity in reproductive-aged women, with rates in the United States steadily increasing over the past decades.¹ In 2022, the prevalence of obesity in women aged 18 to 44 years was 34%, with the highest rates among Hispanic and non-Hispanic Black women.^{2,3} In addition to higher rates of pregnancy complications, such as gestational diabetes mellitus (DM),

preeclampsia, stillbirth, and iatrogenic preterm delivery, several studies have demonstrated the increased risk of labor dystocia and unplanned cesarean delivery among women who were overweight and those with obesity.^{4–7} Of note, 1 meta-analysis found the unadjusted odds ratios of cesarean delivery to be 2.05 for women with obesity and 2.89 for women with severe obesity.⁶ In addition, there are data suggesting that obesity is associated with higher rates of cervical ripening failure during labor induction, requiring a higher total dose of prostaglandins.^{8,9}

The number of patients undergoing elective labor induction has increased steadily after the ARRIVE (A Randomized Trial of Induction Versus

Expectant Management) trial.^{10,11} Over half of labor inductions in the United States require the use of a cervical ripening agent, the most commonly used being misoprostol, a synthetic prostaglandin E1 analogue.¹² Misoprostol can be administered via different routes, including buccal (dissolved in the cheek), vaginal, rectal, oral (swallowed), or sublingual (dissolved under the tongue). Dosing protocols vary across institutions. Both buccal and vaginal administrations prevent first-pass metabolism. Although buccal dosing is associated with a faster absorption rate, it has a lower bioavailability than vaginal administration.¹³

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AJOG at a Glance

Why was this study conducted?

This study aimed to compare the efficacy of buccal misoprostol and vaginal misoprostol for labor induction in patients with obesity by evaluating the number of doses needed to achieve active labor and the mean time to delivery.

Key findings

In patients with obesity, buccal dosing required more doses to achieve active labor, possibly increasing the mean time to delivery. When controlling for other covariates, this difference was not observed with vaginal dosing. Both obesity and buccal dosing were independently associated with requiring more doses for active labor.

What does this add to what is known?

This study adds to the limited body of literature regarding the efficacy of buccal misoprostol for labor induction compared with more commonly studied dosage routes. Based on our findings, vaginal dosing should be the preferred method for patients with obesity.

(IMPROVE) study was a noninferiority randomized trial comparing vaginal misoprostol vs buccal misoprostol in term patients undergoing labor induction.¹⁴ This study suggested that vaginal dosing was associated with a shorter time to delivery, higher rates of vaginal delivery within 24 hours, and lower rates of cesarean delivery for nonreassuring fetal status.¹⁴

As the rates of both obesity and labor induction increase, the optimization of cervical ripening agents in patients with obesity becomes an increasingly important strategy to minimize maternal and neonatal morbidities. This study aimed to determine whether the number of doses required to achieve active labor differed across body mass index (BMI) categories for buccal misoprostol vs vaginal misoprostol in the IMPROVE trial.

Materials and methods

This was a secondary analysis of the IMPROVE trial, a triple-blinded, placebo-controlled randomized controlled trial (RCT) of vaginal misoprostol vs buccal misoprostol in term patients undergoing labor induction (identification number: NCT 02408315).¹⁴ The parent study was approved by the Indiana University Institutional Review Board (IRB). This analysis was exempt

from additional IRB approval. The parent trial was conducted over a 2-year period at 2 hospitals in Indianapolis, Indiana. Participants met the following inclusion criteria: ≥ 14 years of age, undergoing medically indicated labor induction on or after 37 weeks of gestation or elective labor induction on or after 39 weeks of gestation, singleton cephalic pregnancy, and an initial modified Bishop score of ≤ 6 . Patients with previous cesarean deliveries, untreated cervical infections, known major fetal congenital anomalies, or nonreassuring fetal monitoring before the start of labor induction were excluded.

The full trial details are published elsewhere.¹⁴ Briefly, after providing informed consent, the study participants were randomized to receive an initial dose of 25 μg misoprostol, either buccally or vaginally, with a placebo tablet to be given simultaneously via the alternative route. Subsequent doses of 50 μg were administered every 4 hours via the same route per our institution's protocol and in accordance with the American College of Obstetricians and Gynecologists (ACOG) Practice Bulletin.¹⁵ Dosing was continued until either cervical ripening was achieved (defined as a cervical dilation of ≥ 4 cm), additional doses were contraindicated due to tachysystole or nonreassuring

fetal status, or a total of 24 hours of dosing was completed. After the completion of misoprostol dosing, subsequent labor management was per the provider's discretion.

The primary outcome for this secondary analysis was the number of doses of misoprostol needed to achieve active labor (defined as a cervical dilation of at least 6 cm). Participants were grouped into 3 BMI categories reflecting standard classifications for risk stratification (≤ 29 kg/m² [nonobese], 30–40 kg/m² [obese], and ≥ 41 kg/m² [morbidly obese]) based on BMI data collected at the time of labor induction. Descriptive statistics were provided for the demographic data and labor and delivery outcomes by BMI categories. Differences between these BMI categories were analyzed using the Kruskal-Wallis test for continuous variables and the chi-squared test or Fisher exact test for categorical variables.

Labor and delivery outcomes were compared in subgroups of route of misoprostol administration. Poisson regression models were used to determine the differences between the BMI categories for the number of misoprostol doses needed to achieve active labor. Cesarean deliveries were censored at the time of delivery because this could affect the number of doses received. In addition, a model using all dosing routes and BMI as a predictor was developed, adjusting for a combined self-reported race and ethnicity variable and indications for labor induction. All tests were performed at a 5% significance level using SAS software (version 9.4; SAS Institute, Cary, NC). [Supplemental Table 1](#) presents the characteristics of the groupings coded for in the regression model.

Results**Participant characteristics**

Of the 300 participants enrolled in the IMPROVE trial, 299 had complete data for inclusion in this analysis. Within this cohort, 63 participants (21%) had a BMI of ≤ 29 kg/m², 165 participants (55%) had a BMI between 30 and 40 kg/m², and 71 participants (23%) had a BMI of

TABLE 1
Patient characteristics of the cohort

Variable	BMI category (N=299)			P value
	≤29 kg/m ² (n=63)	30–40 kg/m ² (n=165)	≥41 kg/m ² (n=71)	
Mean age (y), mean (SD)	27.2 (5.9)	27.6 (6.3)	29.2 (6.6)	.18
Race				.09
American Indian/Alaskan Native	0 (0%)	0 (0%)	1 (1.4%)	
White	36 (57.1%)	75 (45.5%)	28 (39.4%)	
Black	15 (23.8%)	48 (29.1%)	31 (43.7%)	
Native Hawaiian/Pacific islander	0 (0%)	1 (0.6%)	0 (0%)	
More than 1 race	3 (4.8%)	3 (1.8%)	2 (2.8%)	
Prefer not to answer	9 (14.3%)	38 (23.0%)	9 (12.7%)	
Ethnicity				.02
Hispanic or Latino	19 (30.2%)	56 (33.9%)	11 (15.5%)	
Not Hispanic or Latino	42 (66.7%)	108 (65.5%)	57 (80.3%)	
Prefer not to answer	2 (3.2%)	1 (0.6%)	3 (4.2%)	
Gestational age at trial entry (wk), mean (SD)	39.5 (1.3)	39.2 (1.3)	39.4 (1.2)	.24
Indication for labor induction				.03
Fetal indications	13 (20.6%)	10 (6.1%)	4 (5.6%)	
HTN disease of pregnancy/preeclampsia	5 (7.9%)	41 (24.9%)	19 (26.8%)	
DM/gestational DM	4 (6.4%)	23 (13.9%)	10 (14.1%)	
Postdates/postterm pregnancy	21 (33.3%)	43 (26.1%)	18 (25.4%)	
Elective	8 (12.7%)	19 (11.5%)	8 (11.3%)	
Ruptured membranes/not in labor	0 (0%)	1 (0.6%)	0 (0%)	
Other	12 (19.1%)	28 (17.0%)	12 (16.9%)	
Nulliparous	31 (49.2%)	66 (40.0%)	27 (38.0%)	.36
Mean initial Bishop score, mean (SD)	2.4 (1.8)	2.3 (1.7)	1.9 (1.6)	.28
Route of misoprostol dosing				.66
Vaginal	29 (46.0%)	87 (47.3%)	36 (49.3%)	
Buccal	34 (54.0%)	78 (47.3%)	35 (49.3%)	

Data are presented as number (percentage) or mean (SD), unless otherwise indicated.

BMI, body mass index; DM, diabetes mellitus; HTN, hypertension; SD, standard deviation.

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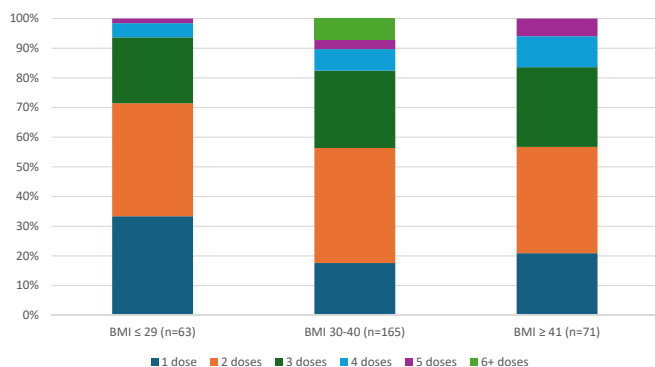
≥41 kg/m². There were no significant differences in age, race, or gestational age between the BMI groups ($P>.05$). There were a greater number of self-identified non-Hispanic participants in the obese (33.9% Hispanic vs 65.5% non-Hispanic) and morbidly obese (15.5% Hispanic vs 80.3% non-Hispanic) groups ($P=.02$). Nulliparous participants were evenly distributed across all 3 BMI categories ($P=.36$) (Table 1).

The most common indication for labor induction was late-term pregnancy (>40 weeks of gestation, 27.4%), followed by hypertensive disorders of pregnancy (21.7%). A greater proportion of participants with obesity or morbid obesity had DM (6.4% in the nonobese group vs 13.9% in the obese group vs 14.1% in the morbidly obese group) or a hypertensive disorder (7.9% in the non-obese group vs 24.9% in the obese group

vs 26.8% in the morbidly obese group) as the primary indication for labor induction ($P=.03$). Participants without obesity were more likely to have a fetal indication for labor induction. A total of 152 participants received vaginal misoprostol compared with 147 participants who received buccal misoprostol. There was no difference in BMI distribution between the dosing groups ($P=.66$). The mean initial Bishop score was 2.2. The

FIGURE

Number of doses of misoprostol needed for participants in each BMI category



The chart displays the individuals in each BMI category who required 1, 2, 3, 4, 5, or ≥ 6 doses of misoprostol to achieve active labor during the study.

BMI, body mass index.

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Bishop scores did not significantly differ between dosing route or BMI ($P=.28$) (Table 1).

Primary outcomes

The overall distribution of the number of doses required for the BMI groups was significantly different ($P<.01$) (Supplemental Table 2 and Figure). Participants with obesity and those with morbid obesity required more doses of buccal misoprostol to achieve active labor than those without obesity (2.1 in the nonobese group vs 2.8 in the obese group vs 2.7 in the morbidly obese group; $P=.01$) (Table 2). Comparatively, there was no difference in doses necessary for active labor when administered vaginally (1.9 in the nonobese group vs 2.1 in the obese group vs 2.3 in the morbidly obese group; $P=.21$). In addition, the mean time to delivery tended to be shorter in the nonobese group that received buccal misoprostol than in the obese and morbidly obese groups (22.7 hours in the nonobese group vs 29.8 hours in the obese group vs 26.4 hours in the morbidly obese group; $P=.07$ using the log-rank test with cesarean deliveries censored). This difference was not observed in the vaginal dosing group (21.5 hours in the nonobese group vs 23.0 hours in the obese group vs 26.4 hours in the morbidly obese group; $P=.49$) (Table 2).

Secondary outcomes

For both the vaginal and buccal misoprostol groups, there was no difference in BMI distribution among participants who delivered vaginally in <24 hours ($P=.91$ and $P=.11$, respectively). The rate of total cesarean delivery and cesarean delivery for fetal distress did not differ between the BMI groups for either dosage form. In addition, oxytocin usage after cervical ripening did not differ between the groups (Table 2).

When adjusting for indication for labor induction and self-reported race and ethnicity, participants with obesity have an expected number of doses higher than participants without obesity (24% higher for those with a BMI of 30–40 kg/m^2 ; $P=.047$; 28% higher for those with a BMI of ≥ 41 kg/m^2 ; $P=.05$) (Table 3). The distribution of the number of doses in the BMI groups is presented in the Figure. When controlling for these same demographic differences and BMI, those receiving buccal misoprostol have an expected 24% higher number of doses than those receiving vaginal misoprostol (incidence rate ratio, 1.24; $P=.005$) (Table 3).

Discussion

Principal findings

In this analysis, participants with obesity required more doses of buccal misoprostol to achieve active labor and

tended to have a longer mean time to delivery than those without obesity (BMI if <30 kg/m^2). However, BMI did not have the same effect when misoprostol was dosed vaginally. Despite this difference, the rates of vaginal delivery within 24 hours did not differ significantly between BMI and dosage groups. Both obesity and buccal dosing were independently associated with requiring more doses for active labor when controlling for other covariates.

Results in the context of what is known

Several studies have compared doses and routes of misoprostol for labor induction.¹⁶ However, buccal dosing remains significantly understudied, so much so that the ACOG currently does not recommend its use due to limited data on its safety and efficacy.¹⁵ A Cochrane Review comparing dosage forms found only 1 study comparing buccal and vaginal dosing, which showed no difference in the time to delivery.¹⁶ However, this study used different doses for the buccal and vaginal groups and, therefore, was not a direct comparison of effectiveness.¹⁷ The parent study to this analysis found that vaginal misoprostol was associated with a shorter mean time to delivery and a higher rate of delivery within the first 24 hours of labor induction than buccal misoprostol.¹⁴ In addition, a RCT comparing buccal misoprostol and vaginal misoprostol in combination with Foley catheter balloon insertion for cervical ripening, found vaginal dosing to be superior to buccal dosing, leading to faster median delivery time.¹⁸

Even fewer studies have compared dosage forms in relation to BMI. In a retrospective cohort study of 1280 patients with obesity (BMI of >30 kg/m^2), vaginal misoprostol was shown to have a significantly shorter time to a 3-cm cervical dilation and to delivery than oral (swallowed) misoprostol.¹⁹ A secondary analysis of the BEGIN trial stratifying patients into 2 BMI categories (<30 and >30 kg/m^2) found that vaginal dosing was more effective for both BMI groups.²⁰ Although the analysis did not directly compare patients

TABLE 2
Labor and delivery outcomes

Variable	BMI category (N=299)			P value ^a
	≤29 kg/m ² (n=63)	30–40 kg/m ² (n=165)	≥41 kg/m ² (n=71)	
Mean doses of misoprostol needed for active labor, mean (SD)				
Vaginal	1.9 (0.9)	2.1 (0.8)	2.3 (0.9)	.21
Buccal	2.1 (0.9)	2.8 (1.0)	2.7 (1.1)	.01
Vaginal delivery in <24 h				
Vaginal (n=89)	18 (20.2%)	50 (56.2%)	21 (23.6%)	.91
Buccal (n=57)	18 (31.6%)	25 (43.9%)	14 (24.5%)	.11
Cesarean delivery				
Vaginal (n=24)	4 (16.7%)	13 (54.2%)	7 (29.1%)	.78
Buccal (n=33)	7 (21.2%)	18 (54.5%)	8 (24.2%)	.95
Cesarean delivery for fetal distress (n=41)	6	24	11	.54
Oxytocin use				
Vaginal (n=100)	15 (51.7%)	58 (66.7%)	27 (75%)	.14
Buccal (n=111)	22 (64.7%)	63 (80.8%)	26 (74.3%)	.19
Mean time to delivery (h), mean (SD)				
Vaginal	21.5 (12.2)	23.0 (12.2)	26.4 (10.5)	.49 ^b
Buccal	22.7 (10.4)	29.8 (12.5)	26.4 (14.9)	.07 ^b

BMI, body mass index; SD, standard deviation.

^a P values were obtained using the Kruskal-Wallis test for continuous variables and the chi-squared test for categorical variables; ^b P values were obtained using the log-rank test with cesarean deliveries censored.

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with obesity with those without obesity, their conclusions were congruent with our finding that buccal dosing is independently associated with a higher number of doses, regardless of BMI.

Although parity was evenly distributed across BMI categories and dosing groups in our study, it remains a key factor in labor induction outcomes. A previously published secondary analysis of the IMPROVE trial limited to nulliparous individuals found no significant difference in the time to active labor or the rate of cervical change between vaginal misoprostol and buccal misoprostol.²¹ These findings suggest that the route of administration may be less influential among nulliparous patients and highlight the importance of considering parity-specific responses in future prospective studies.

Although many studies have evaluated the pharmacokinetics (PKs) of misoprostol dosage forms, most data

exist in the context of first-trimester pregnancy termination, which uses higher doses. Oral dosing and sublingual dosing have been shown to have higher peak serum concentrations than vaginal dosing, in addition to a faster time to peak concentration.^{22,23} In a comparison of buccal misoprostol and vaginal misoprostol for term labor induction, vaginal dosing was found to have a slower rate of absorption, higher bioavailability, and, therefore, higher plasma concentrations than buccal dosing.¹³ This may explain our finding that vaginal misoprostol required fewer doses to achieve active labor than buccal misoprostol due to its prolonged effect.

Clinical implications

With the increasing rates of both obesity and labor induction, understanding the optimal agents for labor induction in patients with obesity is crucial. Based on our findings, vaginal dosing should be

the preferred method for patients with obesity as it may require fewer doses to achieve active labor and is associated with a shorter time to delivery. Although some patients and providers may prefer buccal dosing to avoid repeated vaginal examination during cervical ripening, these findings should be incorporated into the shared decision-making process of selecting a cervical ripening agent.

Research implications

The IMPROVE study included a comparison of PK parameters between vaginal misoprostol and buccal misoprostol. In the study, the addition of BMI as a continuous covariate did not affect the PK model. However, future studies are needed specifically for buccal misoprostol PKs in individuals with obesity to understand the results of our study. Although buccal dosing and obesity were both found to be

TABLE 3

Poisson regression model for number of doses needed for labor induction, adjusting for race/ethnicity, indications for labor induction, and route of delivery

Parameter	Incidence rate ratio	LCL	UCL	P value
BMI (kg/m ²)				
30–40	1.24	1.01	2.68	.047
≥41	1.28	0.99	3.09	.050
≤29	Reference			
Route of misoprostol dosing				
Buccal	1.24	1.17	2.35	.005
Vaginal	Reference			
Race/ethnicity				
Non-Hispanic Black	0.88	0.49	1.16	.20
Hispanic	0.97	0.61	1.43	.76
Other	1.00	0.38	2.65	.99
Non-Hispanic White	Reference			
Indications for labor induction				
Fetal indications	1.12	0.70	2.44	.399
HTN/DM	1.12	0.88	1.88	.188
Other	Reference			
Route of delivery				
Cesarean delivery	0.96	0.58	1.41	.643
Vaginal	Reference			

BMI, body mass index; HTN, hypertension; LCL, lower confidence limit; UCL, upper confidence limit; DM, diabetes mellitus.

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independently associated with the need for more doses for active labor, buccal dosing seems to be more sensitive to the effects of obesity than vaginal dosing. This may be related to higher adipose tissue levels affecting the bioavailability of misoprostol, given that it is a highly lipophilic drug. In addition, the effect of drug receptors, possible genetic polymorphisms, and medical comorbidities, such as preeclampsia, on the absorption, distribution, and metabolism of misoprostol may be investigated. Given the number of labor inductions performed and the inclusion of misoprostol on the World Health Organization's list of Essential Medications, a more complete understanding of the PK of misoprostol in pregnancy, particularly in groups of patients with obesity, is important. In addition, although it was outside of the scope of this analysis, the effect of obesity categories on fetal tolerance of

labor induction should be confirmed with other data.

Strengths and limitations

The primary strength of this study is that it used data from a triple-masked RCT, so neither the provider, patient, nor outcomes assessor knew whether the active drug was being placed buccally or vaginally. In addition, it was a pragmatic trial after cervical ripening was complete, allowing for labor and medication management per provider practice patterns after the trial intervention was completed.

Although this was a relatively large study, the sample size was not initially designed or powered to test for superiority between dosing routes within the BMI strata. Therefore, the inability to detect significant interactions between the BMI category and the vaginal dosage form may be reflective of limited power. The

inclusion of a nonobese comparator group introduces an exploratory aspect to the analysis. Therefore, these findings should be interpreted with caution and considered hypothesis-generating. Validation in larger, prospectively designed studies is required.

In addition, the distribution of participants between BMI categories was unequal, which limits the generalizability of our findings to populations with lower rates of obesity. The largest group was the group with a BMI of 30 to 40 kg/m² ("obese"), which contained nearly twice the number of participants as the other 2 BMI categories. To more closely approximate group sizes, all participants with a normal BMI (<25 kg/m²) and overweight BMI (25–30 kg/m²) were grouped together. The differences in labor outcomes between patients with normal weight and those who were overweight should be evaluated in future

studies. Interestingly, the “obese” group (BMI of 30–40 kg/m²) required the greatest number of misoprostol doses and the longest time to delivery among those receiving buccal misoprostol. We were unable to demonstrate a dose-dependent effect in relation to BMI, which is likely due to our sample size and imbalance in BMI distribution.

Finally, BMI was collected at the time of labor induction. Because of the nature of our patient population, most initial prenatal visits occur well into the second or third trimester of pregnancy. Therefore, prepregnancy BMIs were not available. Given the physiological weight gained during pregnancy, this approach may lead to misclassification of obesity status. It is possible that BMI at term is less representative of body composition and overall metabolic risk of a patient compared with a prepregnancy BMI. In addition, other anthropomorphic indices, such as body roundness index (BRI), may provide a more accurate representation of adiposity, although these were not available in this dataset.

Conclusion

We found that when misoprostol is administered via a buccal route for cervical ripening, participants with higher BMI categories required more doses to achieve active labor. This difference was not observed in the vaginal misoprostol group. Thus, vaginal dosing of misoprostol for labor induction may be the preferred route for individuals with obesity.

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